

Maximal specieslike clusters without a common ancestor

An answer to the CA-free coverage question in S. A. Alexander, *Specieslike clusters based on identical ancestor points* (2026)

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Status. A hand proof by the same method as the paper’s Theorem 9(4). Not machine-checked; no independent review. The literature was not checked for priority.

Source. S. A. Alexander, *Specieslike clusters based on identical ancestor points*, Journal of Mathematical Biology, 2026. [arXiv:2602.05274](https://arxiv.org/abs/2602.05274), [doi:10.1007/s00285-026-02361-x](https://doi.org/10.1007/s00285-026-02361-x).

The question. After Theorem 13 the paper writes: “We do not currently know of any qualitatively different sets of constraints which would positively answer Informal Question 4. We would be particularly interested in a solution not requiring the CA property.”

Result. Replace the common-ancestor property (CA) by a **founding-cohort** property (FC): *every member of the cluster with no parent inside it is born before every member that has one.*

- Together with connectedness, the identical ancestor point axiom (IAP), convexity (CONV) and reflection (REF), FC is closed under increasing unions.
- So every organism lies in a maximal such cluster.
- Maximal FC clusters need not have a common ancestor: a species founded by a *population* keeps its whole founding cohort.
- Any solution without CA must exclude founders who join late (Proposition 4).

Definitions

The paper’s conventions apply (Definition 1: an infinite biosphere G , with parents born before children, finitely many organisms born before any time, and finitely many children each). For $S \subseteq G$:

- a **founder** of S is a member with no parent in S ; the others are **non-founders**;
- **CONN**: S is connected, as an undirected subgraph;
- **FC (founding cohort)**: every founder of S is born before every non-founder of S .

Two facts about FC.

- If S has a non-founder, FC gives S only finitely many founders: they all predate a fixed organism, and G is downward finite.
- If $S \in \text{CONN}$ has no non-founder, S is a single organism: an internal edge would make its head a non-founder.

Relation to CA. Under CONV, CA implies CONN and FC. Every member other than the common ancestor v descends from v , so by convexity its path from v lies in S . It therefore has a parent in S , and v is the only founder. The converse fails; see the example below. Note also that “exactly

one founder" is *equivalent* to CA, since following parents inside S from any member ends at a founder. So allowing several founders is the smallest possible departure from CA.

Let $T_{FC} = \text{CONN} \cap \text{IAP} \cap \text{CONV} \cap \text{REF} \cap \text{FC}$.

Lemma 1. $\text{CONN} \cap \text{FC}$ is upward generic

Let (C_α) be an ascending chain in $\text{CONN} \cap \text{FC}$ with union C.

- **CONN.** Any two members of C lie in a common C_α , where they are connected.
- **FC.** Let r be a founder of C and x a non-founder of C with a parent $p \in C$. Choose γ with $r, x, p \in C_\gamma$. Then r is a founder of C_γ , since it has no parent even in the larger C. And x is a non-founder of C_γ , since $p \in C_\gamma$. FC for C_γ gives $t(r) < t(x)$. ■

Theorem 2. T_{FC} is upward generic

Let (C_α) be an ascending chain in T_{FC} with union C.

- **The easy parts.** By Lemma 1 and the paper's Theorem 9(1, 3), $C \in \text{CONN} \cap \text{FC} \cap \text{CONV} \cap \text{REF}$.
- **IAP.** Suppose $C \notin \text{IAP}$. Then some $v \in C$ has infinitely many descendants and infinitely many non-descendants in C. If C had no non-founder it would be a single organism, so it has one, and therefore finitely many founders. Every member of C is a founder or descends from one: follow parents inside C, which terminates by downward finiteness. By pigeonhole, some founder w_1 has infinitely many descendants in C that are non-descendants of v. Then w_1 is not itself a descendant of v.
- **The König path, as in the paper.**
 - w_1 has finitely many children, so some child w_2 has infinitely many descendants in C that are non-descendants of v.
 - $w_2 \in C$ by convexity (w_1 is an ancestor in C, and w_2 has descendants in C).
 - w_2 is not a descendant of v.
 - Continuing gives an infinite path $Q = w_1, w_2, \dots$ in C of non-descendants of v.
- **The paper's argument now closes, word for word.** Choose α with $v, w_1 \in C_\alpha$.
 - C_α is infinite. w_1 has infinitely many descendants in G, so by REF for C_α it has infinitely many in C_α .
 - Every w_n lies in C_α . Take $\beta \geq \alpha$ with $w_n \in C_\beta$. By REF, w_n has infinitely many descendants in C_β , so by IAP for C_β , C_β contains only finitely many non-descendants of w_n . Since $C_\alpha \subseteq C_\beta$ is infinite, C_α contains a descendant of w_n . It also contains an ancestor, w_1 , so convexity gives $w_n \in C_\alpha$.
 - **The contradiction.** By REF, v has infinitely many descendants in C_α ; it also has the infinitely many non-descendants $w_1, w_2, \dots$ there. That contradicts IAP for C_α . ■

The only change from the paper's proof of Theorem 9(4) is the source of w_1 . The paper takes it from a single common ancestor; here it comes from finitely many founders, by pigeonhole. FC exists to make that pigeonhole available.

Theorem 3. Every organism lies in a T_{FC} -maximal set

For $v \in G$, the paper's Theorem 13 gives $S \ni v$ in $\text{IAP} \cap \text{CONV} \cap \text{CA} \cap \text{REF}$. By the relation to CA above, $S \in T_{FC}$. By Theorem 2 and Zorn's lemma (the paper's Theorem 8), S extends to a T_{FC} -maximal set. Every T_{FC} -maximal set is a specieslike cluster ($\text{CONN}, \text{IAP}, \text{CONV}$). ■

Maximal FC clusters without a common ancestor

The example. Let two parentless founders a (born at time 0) and b (born at time 1) be the parents of every organism of the next generation. Every later organism has two parents in the previous generation. Assume the whole population $P = G$ satisfies IAP, which the paper argues is typical of an unsplit sexual population. For a and b it is automatic: each has exactly one non-descendant in P , the other founder.

- The CA-maximal sets must choose a single common ancestor: $\{a\} \cup \text{desc}(a) = G \setminus \{b\}$, and $G \setminus \{a\}$. Each excludes one founder.
- P satisfies FC: its founders a, b predate everyone else.
- CONV, REF and CONN hold trivially for $P = G$. So **P itself is the T_{FC} -maximal cluster**, containing both founders, and it has no common ancestor.

The immigrant biosphere, where coverage fails without some constraint. In the paper's Fig. 3b, parentless "immigrants" w_i join a lineage at ever later times. FC excludes adding a late founder, so FC behaves like CA there. More generally:

Proposition 4. In the Fig. 3b biosphere, any class of G -subsets that contains every $S_a = \{v_1, v_2, \dots\} \cup \{w_1, \dots, w_a\}$ and is contained in IAP has no maximal element containing v_1 . *Proof.*

- A member containing v_1 and infinitely many w_i fails IAP at v_1 .
- One containing finitely many w_i lies inside some S_β , and every S_β is properly contained in $S_{\beta+1}$. ■

So any CA-free solution *must* exclude founders who join late, which is what FC does. REF and CONV remain necessary, by the paper's Example 15 biospheres (a) and (c): there the chains have a single founder and satisfy FC.

What this leaves

- **Conjecture (S1).** Every T_{FC} -maximal set is $\sim$ -equivalent (the paper's Definition 6: generator sets with finite symmetric difference) to some CA-maximal set, and conversely. The paper's objective-species theorem would then be unaffected by the change.
 - *Evidence.* For a T_{FC} -maximal S with founders R , some founder r has infinitely many descendants in S . The part $S_r = \{r\} \cup (\text{desc}(r) \cap S)$ is in $\text{IAP} \cap \text{CONV} \cap \text{CA} \cap \text{REF}$ and differs from S by a finite set. That follows from IAP at r . It remains to compare S_r with the CA-maximal sets containing it.
- **Biological reading.** FC is a "founding population" convention in place of the paper's "definite starting organism": species begin with a cohort, and no organism whose parents are all outside the species joins it later. Whether biologists find that more natural than CA is outside mathematics.